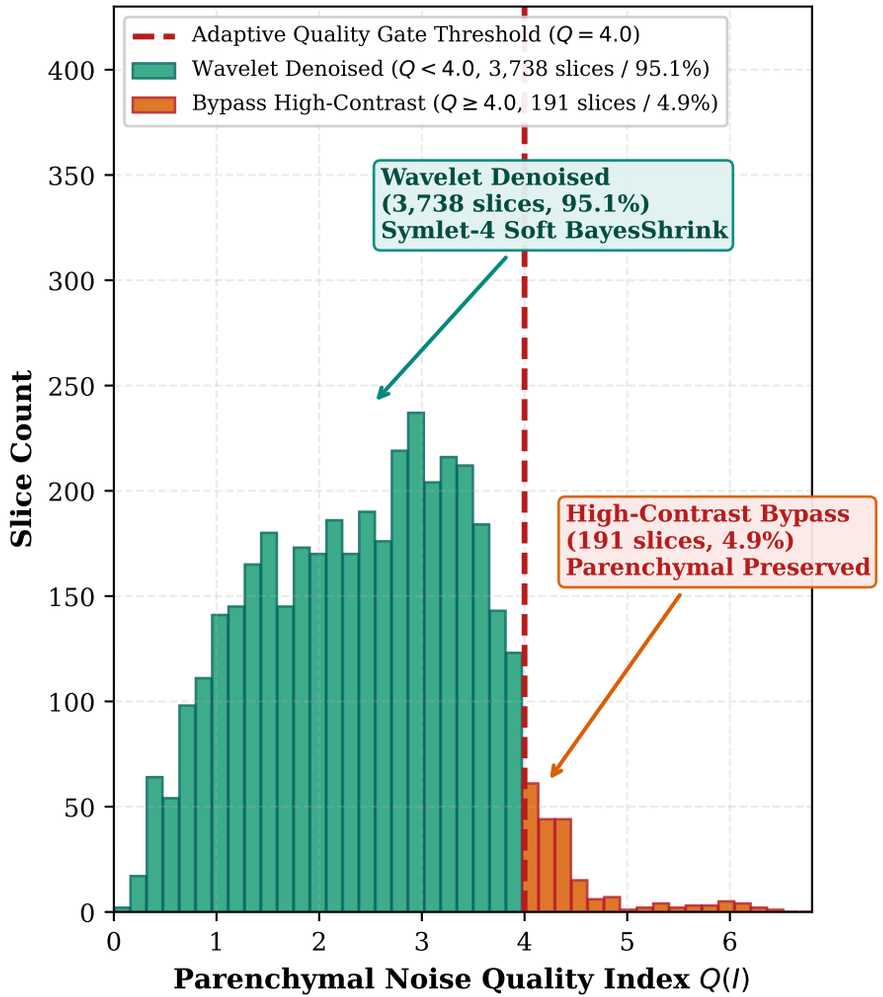
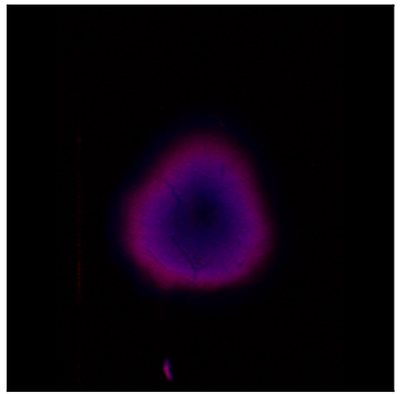
(a) Quality Index $Q(I)$ Distribution & Adaptive Denoising Gate



(b) Preprocessing Audit: Excluded Artifact Slices (Top) and Quality-Aware Wavelet Gate (Bottom)

Artifact 1: FOV Truncation



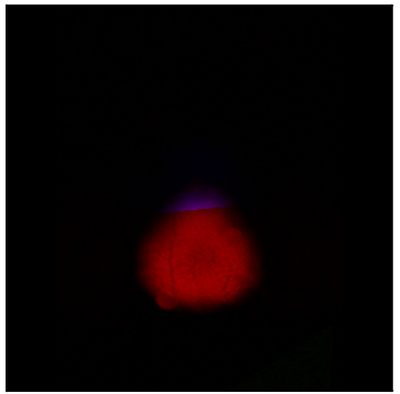
$ED = 0.0023$, $CR = 0.0020$
$d_c = 0.64$ | **Excluded (0 px)**

Artifact 2: Scanner Noise Fringe



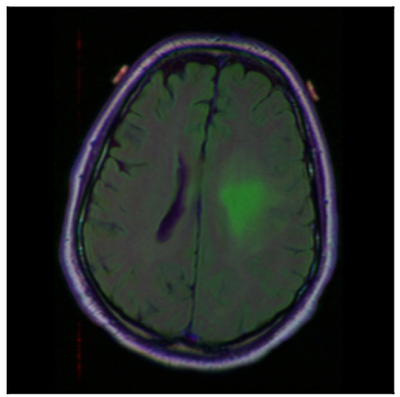
$ED = 0.0029$, $CR = 0.0024$
$d_c = 0.32$ | **Excluded (0 px)**

Artifact 3: Non-Parenchymal Fringe



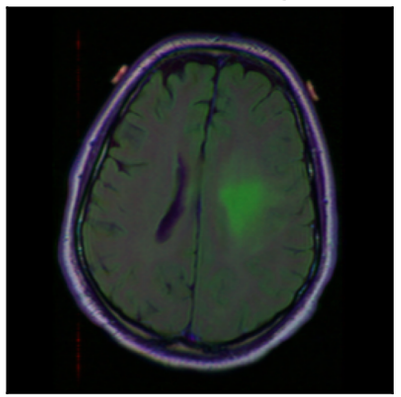
$ED = 0.0000$, $CR = 0.0000$
$d_c = 1.00$ | **Excluded (0 px)**

Raw Slice (Moderate Noise)



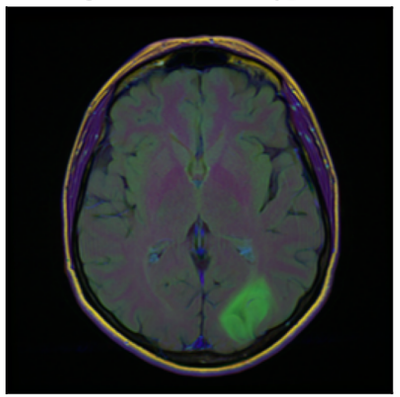
$Q(I) = 2.37 < 4.0$
Raw Input Slice

Wavelet Denoised (Symlet-4)



BayesShrink Denoised
Artifacts Suppressed

High-Contrast Bypass



$Q(I) = 4.12 \geq 4.0$
Preserved (No Denoising)